

Pizza Saucer

by TN: Team Name

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IGME-601 Game Development Processes

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This should update correctly whenever refreshed if you use the formats already defined consistently.

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Executive Summary

You've been wandering the galaxy for months with no luck. With your navigator down, you've flown into countless asteroid fields and space wars. You don't know the way home... but you know how to make pizza.

Satisfy your alien customers, discover new ingredients and recipes, and find your way back home!

High Concept

Cooking intergalactic pizza for others evokes a sense of comfort, community, discovery, and connection.

Production

Who is on the core team, and what are everyone's primary roles & responsibilities? What responsibilities are shared by everyone? What are the major risks & personal goals that influence your production process?

All teammates will be game designers and assist with the narrative.

Gia: producer, programmer

Kaijie (KJ): artist, background and asset designer

Savannah: programmer, narrative designer, sound/music curator

Shaolin: programmer, source code control manager

Wenpeng: artist, character designer

Target Audience

Cozy gamers who enjoy food, sci-fi, and narrative mysteries.

Design Influences

VERY briefly, what are the primary inspirations/references you'll be using? How/why? (You'll detail these more in later chapters as your research continues throughout the semester.) These could be similar games, games that do something similar, or assets that capture the aesthetic of your idea. Consider references that aren't even games.

Cooking mechanic: Venba, Good Pizza Great Pizza, Papa's Pizzeria

Primary style: We aim to adopt the style of Hollow Knight for the overall game, though with more vibrant colors and brighter lighting to create a lighthearted, casual atmosphere.

Game Design

Design Pillars

1. Satisfaction: players use clues to figure out what kind of toppings to use, and reach a satisfying conclusion when the alien enjoys their meal. The player will also be trying to satisfy their alien customers.
2. Discovery: learn about different alien species, ingredients, ingredient interactions, planets, and alien preferences.
3. Connection: learning about the world and the aliens through cooking

Core Gameplay

Briefly describe what the game is from a gameplay and/or story perspective. For each item, give it a short description and how it relates to one or more of the design pillars listed above. If an item doesn't relate to a design pillar, either your pillars need to be adjusted, or this item isn't vital for this game.

- Cooking: putting ingredients together on the pizza
- Recipe book: finding information on ingredients and preferences
- Exploration: flying ship to different planets and meeting new aliens
- Mystery: collect clues to fill your map and find a way home

Aesthetics

Briefly describe the overall aesthetics of your game. How does this support your design goals?

Our game features a simple, cute art style with light touches of fantasy. For character design, we plan to adopt a simpler, more casual cartoon style for the alien NPCs. This approach maintains a consistent game style while ensuring the game's aesthetic appeal won't bore players.

Technical Approach

- Engine: Unity (C#)
- Version control: Github
- File management: Google Drive
- Task management: Codecks
- Communication: Discord, weekly meetings

Current Status

We just completed our final sprint and have published our game to itch.io. In this final stretch, we added several things to add to player experience including sound effects, more dialogue options, ingredient interactions (with recipe book updates), a tutorial, and cutscenes.

Production

*During the span of this project, you are a game studio. Describe your production culture, communication and pipeline. This chapter drills down on your current process. **As it changes, update this!!** You should have the 1st pass of everything here complete by the end of sprint 1 and then continue to update it throughout the project.*

Team Organization & Communications

How did you structure your team? What skills did you have available as a combined group and how did that influence your objectives and process? How will you all communicate? What is a reasonable response time expectation? What are the team's overall priorities? How will you handle disagreements? How, as a TEAM, will you make sure everyone has value-added work to do and has a "voice" on the project?

All teammates will be game designers and assist with the narrative.

Gia: producer, programmer

Gia studied computer science in undergraduate with a minor in computer gaming. She has previous experience on two projects acting in a producer role.

Kaijie (KJ): artist, background and asset designer

Studied illustration.

Savannah: programmer, narrative designer, sound and music curator

Savannah also studied computer science with minors in creative writing and studio arts.

Shaolin: programmer and source code control manager

Wenpeng: artist, character designer

As a group, we have shared values of hearing everyone out and contributing meaningfully to the team. We also want to acknowledge and appreciate other's work. To help with this, we have a progress channel in our Discord where team members can send any of their work, whether finished or in-progress. We also collectively prefer to handle conflict through discussion. Our Codecks helps us to distribute work equally and make sure everyone is contributing. Although tracking hours is important, we are placing more of an emphasis on equal effort.

Resource & Risk Analysis

What skills and roles will you need? Be open to additional team members, keeping in mind that you probably will have more work than you're expecting!

What technology will be required throughout the project, in terms of hardware, software, materials, etc? You might also want to include monetary costs if you will be working in the analog world and expect to do any actual production and purchasing.

Also DISCUSS the major risks your team will have/had to account for during the project. For each, cover:

- *Likelihood - What were the chances of this happening?*
- *Severity - What would the impact have been?*
- *Mitigation - How would/did you respond/minimize the impact of this?*

This is not simply a table with ratings (although formatting as a table can be helpful). Discuss each and how it influenced your decisions and the project overall.

We do not expect to need additional team members for this project. We will also not need any analog materials.

Our team is relatively balanced, with 2 artists (character and background) and 3 programmers. Our programmers all are comfortable with software development, source code control, and working in Unity. We are most inexperienced with overall game and level design as well as music and sound production.

Wenpeng Liao	Team 1	9:00am - 01	1	2	2	2	1	2	2	3	2	3	3	3	2	2	2	2
Kaijie Yang	Team 1	9:00am - 01	0	1	0	1	0	1	1	3	1	3	1	2	0	2	2	2
Gia Hur	Team 1	Gia Hur P.C. 9:00am - 01	3	3	2	2	3	2	2	3	2	3	1	1	0	1	3	2
Savannah Balistreri	Team 1	9:00am - 01	1	2	2	2	2	3	1	2	2	3	2	2	2	1	2	2
Shaolin	Team 1	9:00am - 01	3	3	3	3	3	3	2	3	1	2	1	2	0	2	1	1

Risks	Likelihood	Severity	Mitigation
Less experience with game design	High	High	We are all working together with our game design. In addition, our gameplay is relatively simple in terms of level design.
Sound and music production	High	Low	We will source our sound and music. We may try to produce sounds.
Language barrier	High	Low	We have language barriers in our team. We mitigate this by speaking clearly and using translators when necessary.

Priorities

Describe the overall priorities and rationale that will guide your development plans. While the detailed scope plans may change, these overall priorities should be more stable. If, via playtesting & research, you discover your guiding priorities do need to change, make sure to discuss it with your faculty!

Our top priority is incorporating a mystery element. Our initial idea came easily, as our VICKY revealed common interest in cooking and sci-fi. We had more difficulty including a mystery element. We also want to make sure that clues are structured in a way that challenges

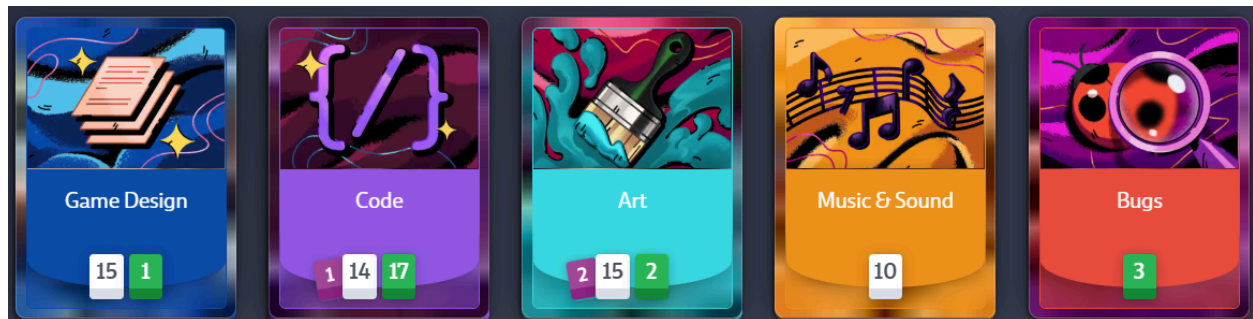
players. We also want fun gameplay. We are concerned that players will get bored. We will focus on this for our playtest feedback forms.

Task Management

Given the priorities and target scope + your team organization and communication strategies, how are you tracking what needs to be done? How do you know who is doing what, when? How are dependencies managed and progress assessed? Describe this from both strategic (the overall model: overall workflow, etc.) and tactical (how you enacted your process strategy: task management procedures, etc.) perspectives. Be specific and include your rationale for these approaches and their impact on the final product quality and how your priorities and risks motivated your process decisions. (Use an appendix for detailed task management procedures, etc. as needed to keep this section from getting cluttered.)

We use Codecks and Discord to communicate tasks. Codecks is a project management software specifically designed for game development teams. It includes decks (game design, code, art, music and sound, bugs), time tracking, effort, importance, metrics by week and deadline, etc. All group members can add cards (tasks), but most are added by the producer. Each member can look at their hand to see tasks assigned to them. Members are expected to mark their own card effort, importance, and completion.

We also use a built-in Discord bot that notifies the team when cards are created, started, completed, etc.



Version Control

Summarize how you're storing/managing ALL product artifacts (not just code!). Why are you using these approaches/tools? Are there key features of the tools you are leveraging? (e.g. pull requests).

We are using Github for code. Shaolin is the source code control manager. We use separate branches and use pull requests when merging into main. We also have technical documentation to explain each class.

We use a shared Google Drive for documents and art assets. The Table of Contents document in the Google Drive clearly lays out how to name each asset for easy management.

Asset Pipeline

Summarize how you store, manage, and incorporate assets into your game. Include any strategies for coordinating with external team members here as well.

All assets are drafted, created, and stored in Google Drive by our artists. Our programmers then download these assets to incorporate into the game. Programmers

must indicate using Google folder colors whether the assets have been implemented (green = implemented).

Audience & Market

Target Audience

Who will play your game? Who will buy your game? How will the game design & your priorities/production process meet their needs? -- From the perspective of attributes of the audience and your game. Define the audience and how to address their interests/needs using relevant references.

We are targeting cozy gamers who enjoy cooking games. We want to prioritize an interesting puzzle mechanic, where players infer customer preferences from their appearance. The game is set in an alien world where the player regularly interacts with their alien customers. Under this gameplay is a story of the chef trying to get home. Our art style is cozy and cute, so this game is welcome to all ages.

Competitive Analysis

Now that you know who you want to target and what they want, what defines this space in the market - ESA reports, other games, ... How is our game like this, and how is it unique. Who will buy your product? How well have similar products sold? Justify your sales estimates. How will your idea/innovations generate sales?

Pizza cooking simulator games are extremely popular. Typically, people think of old flash games such as Papa's Pizzeria. At this time, the biggest game in the market currently is *Good Pizza, Great Pizza*, a mobile application by TapBlaze. In this game, you fulfill pizza orders while trying to make a profit and keep your business running. There are several storylines throughout the game including rival pizza businesses and wacky customers. This game has 100M+ downloads on Google Play and 250M+ on Steam.

Our game is similar in mechanics, as it involves cooking pizza, unlocking ingredients, and using a recipe book. *Good Pizza, Great Pizza* is also a cozy, narrative-based game. However our game is unique because of the space setting. This brings in exciting elements such as alien customers, ingredients from different planets, and unfamiliar planets. We also weaved in a plot of conflict among the alien species as the player tries to find their way home.

Marketing & Distribution

How will you market your product? How/where will it be distributed?

We have posted our game on itch.io. Previous sprints were hosted on Unity's WebGL Build server. We will have both an OpenGL build for players to easily access on their web browser and a Windows build. We will advertise our game to other IGME students through Discord, as well as in class Discords. We will also be sending out our playtest form to receive any feedback.

Design Influences

Then, given your audience + the overall project goals and game design defined in the executive summary, do a detailed analysis of existing games/media. How is each similar to/different from what you are trying to accomplish? What worked well? What didn't? What can you leverage/learn?

Consider this from multiple perspectives! Some suggested subsections are below, but you'll likely have to add/modify these based on what is relevant to your project! It can also be helpful to add a table to each section to make it easy to compare games. For each area, you should do the following:

- *Compare the common (and uncommon) media in this "space." Ideally, you should have a table that notes key elements of each inspiration as well as how your game will compare.*

Add subsections as needed to break this up in a relevant way to support your game! For example: genre, specific design pillars, core mechanics, onboarding approaches, etc.

Cooking Mechanic

Venba is a cooking game, focused on the story of an Indian mom who immigrates to Canada. You cook generational recipes and learn more about growing up as an immigrant. We liked the top-down cooking style and the drag-and-drop ingredients.



Good Pizza Great Pizza is a mobile pizza cooking simulation game. There are several storylines to progress through as you interact with customers and rival pizzerias.



Papa's Pizzeria is a flash game in a series of other Papa's games. We like the game's simple mechanics and style.



Game Design

Add concept mockups and, eventually, game screenshots throughout this section.

*This is the big picture - not just what you implemented. **Include WHY you made these decisions**, not just a report on what the game is. This is not a user manual!*

Explain your game to potential players, investors, etc. - NOT to developers. Yes, describe your mechanics, but do it in the context of the game concept, theme, etc. All of this should be specific to YOUR game. For example, for a 2d platformer, this chapter shouldn't be applicable to any platformer, just YOURS.

Details about how mechanics are implemented, etc., go into technical design. This is about the objectives of mechanics, level design, etc.

Add subsections as needed to break this up in a relevant way to support your game! For example, setting & narrative design, theme, core gameplay loop, mechanics, UI, onboarding, etc.

Summary

- First-person 2D
- Setting
 - Alien world
 - Alien customers
 - Alien ingredients
 - More unique character design
- Story (mystery)
 - On planet, guidance machine broken
 - Figuring out how to get home
 - Kitchen rocket ship
 - Map in restaurant with all planets
 - Chef draws map with safe channels
 - Wrong path will kill the player
 - End goal
 - Go home
- Cooking
 - Pizza
 - Various planet types
 - Recipe book gives basic descriptions of ingredients

Gameplay

- Drag and drop toppings onto pizza
- Aliens give hints to their preferred pizza
 - First few aliens will have simple/straightforward orders
 - Clear hints / stating what toppings they like
 - Later aliens will be more vague, only saying flavor/texture/color preference

- Serving the correct order makes the alien happy: they will give you information on their planet / the map
- Using this information, players will fill out their map, drawing paths to the various planets
- Connecting to a planet and clicking transports the player to a new planet with new aliens and recipes
- Connecting to the home planet ends the game

Mindset

We want the player to feel comfortable while playing the game. While there is a mystery element and puzzles, the overall gameplay is calm and simple. We hope that solving the puzzles will not be too difficult, and will give the players satisfaction.

Storyline

We want the player to feel invested in the characters of our game, and the worlds they explore. In order to achieve this, we crafted a storyline with conflict that the player could see firsthand through the windows of their kitchen, and help resolve with their cooking.

As our team's shared interests included sci-fi, the storyline was heavily inspired by media like Star Trek and Star Wars. Wars throughout space are a common trope, and we wanted to play with that with our pizza theme.

A large point of difficulty for our storyline is that it not only has to advance the story of the game, but it additionally needs to serve purposes for the mechanics of our game. Our dialogue as a whole needed to:

1. Advance the story.
2. Give hints towards what toppings aliens wanted on their pizzas.
3. Allow the players to discover planets.
4. Allow players to discover paths between those planets.
5. Give hints as to whether those paths were dangerous or safe.

Because of this, it was difficult to work out what our storyline would be based on all these struggles, but after the addition of our glitch aliens, we decided to have our story be about a space war with a twist.

As the player travels to more planets and talks to more aliens, the player will hear more and more about the glitch planet and their species. They find out quickly that being touched by a glitch alien turns you into one, and that most aliens would rather die than be infected. This has led to space wars in order to rid the galaxy of the glitched aliens. The information the player receives slowly leads them from planet to planet until they reach a point where there is no other planet they can visit, besides the error planet. Upon visiting the glitch planet, the player learns that the glitch aliens are

not wanting to hurt anyone, and that the planet consists of aliens from other planets that have been infected. The player, by talking to these glitched aliens, finds where home is and is able to travel back home - and all the aliens, including the glitch, come to visit the player after they get back to earth, effectively marking the end of the space war.

This is a play on the space robots trope in many space movies and tv shows. One notable space robot plot is the Borg in Star Trek, where the Borg, in an attempt to take over the entire known universe, infects living creatures to turn them into one of the many Borg in their hive. With our plot, we played with this notion of infection, but added the twist that the glitch aliens don't want to infect others - they just do. It's up to the player to discover this and show that the glitch aliens are safe, and the best way to do this is by serving pizza.

We noticed the players would try to travel without speaking to the aliens. To facilitate player connection with the aliens, we added a reveal mechanic. This means that once a player speaks to an alien and gains information on a planet, this planet will appear on the map. Then, as they are given information on safe and unsafe paths of travel, the player can mark these on their map.

Now, since players are forced to interact with the aliens and discover things about them through the dialogue, our design pillars of satisfaction, connection, and discovery will shine through our narrative and associated mechanics.

Aesthetics & Assets

Since the complete asset list might be quite large, move it to an appendix and provide just an overview in this chapter. Use this chapter to provide examples (& sources!) and address how the assets serve your desired game design goals.

*For each, discuss the overall goals and how you made decisions. It provides background for readers but also sets up how to make decisions when collecting assets for future development. **Imagine what you would tell a content creator about expectations for creating assets to fit the game design.***

How did you approach/address the soundtrack, audio feedback, character animations, environment, feedback animation, and UI -- those are the basic buckets.

Add subsections as needed to break this up in a relevant way to support your game! For example, environmental assets, animations, vfx, sound effect, music, etc.

Style Influences

Primary Aesthetic



Character attire and overall proportions are as shown below:



We aim to adopt the style of Hollow Knight for the overall game, though with more vibrant colors and brighter lighting to create a lighthearted, casual atmosphere.



Hollow Knight game link:

https://store.steampowered.com/app/1030300/Hollow_Knight_Silksong/

We want the game's overall style to avoid making players feel tense about the unknown, instead using a more welcoming approach to help them focus on gameplay and exploration.

Our music influences for Pizza Saucer are Charlie's Here from the pizza parlor minigame in Club Penguin, and the theme song from Neko Atsume. Both songs are calming and can be listened to repetitively, but when making our music for the game, we want it to be more sci-fi in style, while still retaining the simplicity from the Neko Atsume theme and the jazzy tones from Charlie's Here.

[Neko Atsume Theme](#)

[Charlie's Here - Pizza Parlor Club Penguin](#)

For the SFX of our game, we were inspired by real world noises, like cheese slapping noises for dropping toppings on the pizza, and for our character voices, we were inspired by a meme video and the character Glep from smiling friends, linked below.

[Glep from Smiling Friends](#)

[Meme Video](#)

Style Attributes

Our game features a simple, cute art style with light touches of fantasy. This approach saves production time and reduces unnecessary workload while still creating an inviting and engaging atmosphere. The whimsical yet approachable look supports our design goals by drawing players into the world quickly, making the mystery elements more appealing without overwhelming them with excessive visual complexity.

For character design, we plan to adopt a simpler, more casual cartoon style for the alien NPCs. This approach maintains a consistent game style while ensuring the game's aesthetic appeal won't bore players.

Asset List

[Asset Master Doc](#)

Sounds & Music

[Sounds Master Doc](#)

Technical Architecture

This is your technical design document. No code should be in the document unless there is something incredibly clever/core to the implementation of the game.

Please do NOT go overboard. This is NOT an onboarding doc for new developers - it's your overall architecture and why you made the decisions you did. Reference external sources as needed. LOTS of pictures!

Focus on models that might be helpful and the level of discussion that helps readers know how to interpret the model. Super detailed models are generally useless. Ensure the detail level matches the discussion depth in each section. Lots of small models are much more helpful.

*Even in technical design, the language here should be in the context of your game! Avoid jargon, etc. As said by a past IGME-601 student, **frame everything in the context of "Architecture in Service of Gameplay"**:*

I want to see how each team considered the architecture of the game and how it could be best designed to deliver gameplay that lent itself to the larger vision of the game.

- *Discuss the evolution of the code and how the design decisions influenced gameplay.*
- *When the reasoning for decisions is discussed, explain how this architecture is best suited to make the game fun for players and show off their skillsets vs. solely referencing ease of development and implementation (which is still very important!)*

Without this, how the design choices should best translate to technical design isn't obvious, so knowing how best to implement any given mechanic might be a personal judgment call rather than something derived from the game design.

Focusing on how the technical aspects are in service of the game itself:

- 1. Motivates why this technical work is important.***
- 2. Keeps the non-technical members of the audience engaged (and even the technical ones - because tech alone is really dry).***

Screens

1. Front House
2. Back Kitchen
3. Map UI
4. Recipe Book
5. Start Menu

Controls

We want to have all mouse controls for our game.

Dialogue

SPACE/LMB: advance dialogue

LMB: choose dialogue option

Cooking

LMB drag: bring pizza to oven, conveyor belt, or trash can

LMB drag: serve pizza to alien

LMB: toggle pick up ingredient, place ingredient

LMB bin again to empty hand

RMB: empty hand of ingredients

Map

LMB drag: draw paths to planets

LMB: travel to connected planets

RMB drag: erase paths, draw unsafe paths

Recipe Book

LMB: Click to see previous/next pages

Switch Screens

Hover over mouse icon

Aliens

Scriptable Object

Contains important information for each alien including name, happy prefab, sad sprite, and pizza preference. It also includes dialogue pools including the greeting pool, clue pool, and casual pool. Each of these also contain different weights.

Script

Contains behavior to handle the alien entering and leaving, receiving a pizza, and acting accordingly (happy, sad). This triggers the happy or sad sprite, the dialogue response, and sound.

Audio

Central system to manage sound effects and music. Supports audio playback through strings, lookup, and global access. Assigning this script to any GameObject allows playing sound.

CinemachineCameraController

Changes Cinemachine camera's priority to switch which camera is active in the scene.

CSV reader

Loads and parses CSV files to handle the dialogue system.

Dialogue

Dialogue Manager

Starts and stops Yarn dialogues based on title and stores prerequisite checks. Also deals with special cases such as pausing dialogue during the tutorial and before the alien sits down.

Scriptable Object

Stores dialogue metadata such as title and prerequisites.

Level Manager

Handles all kitchen and front-house gameplay:

- Spawns aliens using planet data
- Tracks player location
- Loads and clears levels based on travel
- Unlocks new toppings on new planets
- Plays kitchen audio
- Manages oven, alien seats, and alien spawning
- Triggers game over screens

Map

MapManager

Manages player's current planet, all planets on map, all connections made by player, route safety, travel requests, transitions (to new planet or to death screen), handles map UI interactions. Also handles scene loading for final cutscene.

MapUI

Handles all visual interactions on the map. This includes displaying planets and routes, erasing, and sound effects.

PlanetUI

Visual representation of the planet. Shows outline and name on hover, travels on LMB (if connected to current planet), provides position to LaneUI.

Lane

Represents the path between two planets.

Lane UI

Draws connection on map with UI element. Drawn with left drag and removed through right drag.

TravelConfirmUI

Popup to confirm player travel. Displays the destination and confirm button. Triggers travel via MapManager on confirmation.

MapSnapshot

Captures image of map to display at front of house as UI element.

Oven

Oven

Plays sound when turned on, stops when turned off.

MeterFillManager

Displays cooking progress (green for cooking, red for burning). It plays sounds such as ticking while cooking, “ding” when perfectly cooked. Resets UI when pizza is removed, and tracks pizza cooking state.

OnboardingManager

Tracks player actions during onboarding to properly trigger next dialogue. This includes:

- Entering kitchen / front-house
- Placing dough on table
- Picking up toppings
- Adding toppings to pizza
- Returning pizzas (having an empty hand)
- Cooking pizza
- Serving pizza
- Opening map
- Marking a planet as dangerous

Pizza

PizzaManager

Handles pizza behavior. This includes generating a pizza base when dough is clicked on and ensuring the counter is not occupied. It also tracks whether a pizza is in the oven to ensure there is no overlap. It tracks whether the player is holding a topping, managing the conveyor belt spawn points and speed, and cooking durations.

Pizza

Stores pizza base, holds toppings with quantity, tracks cooking time and cook state (raw, cooked, burnt), checks if order is correct, reports incorrect elements (missing, unwanted, or insufficient toppings).

Scriptable Object

Has predefined pizza recipes, converts data into pizza instances.

PizzaBase Scriptable Object

Registers pizza base into a global dictionary and creates sprites.

PizzaInstance

Interactable pizza in the kitchen. Handles drag and drop interaction (collision with oven, aliens, conveyor belt, and trashcan).

In oven: handles cooking states, changes color while cooking.

Toppings: handles placing toppings.

Handles moving on conveyor belt to front house.

Planets

Planet Class

Simple wrapper class that stores PlanetSO.

Scriptable Object

Contains planet name, highlight color, background sprite, unlocked pizza toppings, and aliens found on the planet.

Recipe

Book Contents

Handles known recipe pages, UI for turning pages, showing entries, and opening the book. Also responds to Yarn commands to reveal interactions.

Recipe unlocking

Creates a new page for unlocked topping and adds to the book.

Toppings

ToppingBin

Handles toppings being picked up, dragged, and placed. On click, the player picks up and returns toppings (or on right click). On pickup, a “ghost” topping is shown, following the mouse. When over the pizza, it places the toppings with a sound effect. It then reports these to the onboarding manager.

Scriptable Object

Defines topping type, sprites, prefabs for bin, and registers everything into a global dictionary.

Overview

Discuss, in plain terms, your overall approach to technical design for the game. Think of this as a summary of the rest of this section. You should probably include a diagram of major subsystems and any other important information needed to understand the structure of your technical design. Address how its design serves your desired game design goals. Make sure to cover the fundamental choices you made in the game's development process. Start with a big-picture view, then organize/describe your technical approaches and decisions.

Game Production Schedule

Sprint 0	9/14 High Concept and Initial Project Approach
Gia	Production (Task management, team doc, schedule meetings, team status), brainstorm, recipe book mechanic
Kaijie	Brainstorm, find references for background and asset art, planet sketches
Savannah	Repo setup, brainstorm, cooking (drag and drop mechanic)

Shaolin	Brainstorm, map mechanic
Wenpeng	Brainstorm, find references for character art, alien sketches

Sprint 1	9/15-10/5
Gia	Production, recipe book mechanic (open, close, flip page, serialize recipes), dialogue system
Kaijie	Kitchen design, planet sketch
Savannah	Cooking (drag and drop, change pizza sprite, serve alien, alien reaction)
Shaolin	Map mechanic (open, close, snapshot, draw paths, draw non paths, travel to connected planets)
Wenpeng	Alien design, food design, map background

Sprint 2	10/6-10/26
Gia	Production, Recipe Book Updates, refactor serving pizza
Kaijie	Refactor serving pizza Travelling to other planets, connect to narrative
Savannah	Recipe book Dough Toppings in bins
Shaolin	Rest of 6 toppings Fix background
Wenpeng	Narrative (alien hints to home planet) Game Flyer

Sprint 3	10/27-11/16
Gia	Bug Fixes - Dialogue and Oven, Tutorial
Kaijie	Planet, Background, and Alien Food Art
Savannah	Narrative and Dialogue Overhauls, Tutorial
Shaolin	Bug Fixes - Pizza and Aliens, Tutorial

Wenpeng	Cutscene Art
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Sprint 4	11/17-12/8
Gia	Production, bug Fixes
Kaijie	Background Art, Planet Art
Savannah	Tutorial dialogue, bug fixes, sound
Shaolin	Bug Fixes
Wenpeng	Itch.io art: cover image, banner

Play Testing

We received a lot of valuable feedback in our playtests, as logged in our sprint reflections document. Below we have detailed specific notes made by playtesters that we decided to implement (or not) and why. For the purposes of this document, we have excluded bugs.

Feedback	Response
Playtest 1	
Players reported a $\frac{1}{2}$ difficulty, suggested having multiple toppings	We agreed that having specific recipes for each alien would add some difficulty to keep the game interesting without being too hard.
Players can win without playing by guessing the correct path	We will implement a reveal mechanic where planets will only reveal and paths will only be travelable after specifically mentioned by an alien.
Playtest 2	
Switching between the kitchen and front was difficult	We considered shortcut keys, but wanted mostly mouse controls. We also considered having the screen immediately change to the front when the pizza is placed on the conveyor belt. Ultimately, we decided to have the screens switch if a player hovers over the area of the screen.
Confusion around the narrative	We decided to add onboarding, a tutorial, and an introduction cutscene

Dialogue is repetitive, can get boring when interacting with the same aliens	We will now use dialogue pools. Aliens will not repeat dialogue (unless a player continuously gets their order wrong), and will have more “casual” dialogue options rather than always giving clues.
Placing toppings can be tedious	Professor Negrin suggested switching the controls so that when you click on an ingredient bin, you can just click on the pizza to place the toppings until you unclick.
Players had trouble determining when the pizza was cooked	Oven ding sound Oven meter
Playtest 3	
Controls confusing (switching screens, placing ingredients, oven)	We need to add a tutorial that explains all controls to the player.

References

Give credit to ideas that are not yours and the games that influenced you in ALL written work. Use APA style, which is standard in the Learning Sciences disciplines.

Recommendation: use Zotero free bibliographic software to automate your citations. It allows you to store citations for future reference and automatically reformats to different styles. Note that Zotero APA still requires capitalization correction to “sentence case” as opposed to “title case” - https://www.zotero.org/support/google_docs

*Use the following page for learning in-text citations and how to write a references section:
https://owl.purdue.edu/owl/research_and_citation/apa_style/apa_style_introduction.html*

For games, cite designer (as author), year, and publisher using APA style – choose one of these:

- <http://askus.library.wvu.edu/faq/116850>
- <http://moonflowerdragon.blogspot.com/2010/03/how-to-cite-game-in-apa-style.html>
- <https://www.bibguru.com/q/apa-video-game-citation/>

If a discussion is based on something that is not your original idea/work product, you must include a citation!